Machine Learning for Radio Pulsar Identification

HTRU2 Dataset | Reference Reproduction and KNN Extension

1. Problem Statement and Dataset

Problem. Radio pulsar candidates must be distinguished from non-pulsar signals. Since pulsars form a small fraction of the data, a model that predicts mostly non-pulsars can achieve high accuracy while still missing real pulsars. The project therefore evaluates both overall correctness and pulsar detection.

Dataset. HTRU2 contains 17,898 samples, eight numerical features derived from integrated pulse profiles and DM-SNR profiles, and a binary class label. The dataset has no missing values.

Dataset property	Observed value	Significance
Non-pulsars (class 0)	16,259 (90.8%)	Majority class
Pulsars (class 1)	1,639 (9.2%)	Minority class
Train / test split	14,318 / 3,580	Held-out test evaluation
Training after oversampling	13,007 per class	Balances training classes

Procedure. An 80:20 split was used. Oversampling was applied to the training data only; the test set retained its original class distribution (3,252 non-pulsars and 328 pulsars).

2. Existing Approaches and Proposed Add-on

Part	Method	Purpose
Part I — reproduction	Gaussian Discriminant Analysis (GDA)	Probabilistic supervised baseline
Part I — reproduction	Random Forest	Ensemble classifier; threshold can tune recall/precision
Part I — reproduction	K-Means	Unsupervised grouping of pulsar samples
Part I — reproduction	TensorFlow SOM	Maps pulsar samples onto a 5×5 neuron grid
Part II — extension	K-Nearest Neighbors (KNN)	Additional supervised model; select k using cross-validation

Proposed/add-on approach. KNN was added as the extension model beyond the reference study. Candidate k values were evaluated with five-fold cross-validation, and k = 3 was selected. PCA was used to visualize high-dimensional clustering results in two dimensions; it is a visualization step, not a classifier.

3. Part I — Reproduced Results

Supervised models

Model / evaluation	Accuracy	Precision	Recall	F1	ROC-AUC
GDA — test	97.85%	87.24%	89.63%	88.42%	0.9708
Random Forest — threshold 0.5	98.02%	91.86%	85.98%	88.82%	0.9661
Random Forest — threshold 0.2	97.49%	82.69%	91.77%	86.99%	0.9661

- Cross-validation: GDA = 97.52% accuracy and 87.49% recall; Random Forest = 97.92% accuracy and 86.42% recall.
- The paper reports GDA test accuracy/recall of 96.28%/83.43%. Our values are higher. Random Forest cross-validation is close to the paper's 98.02% accuracy and 85.31% recall.
- Threshold effect: reducing the Random Forest threshold from 0.5 to 0.2 raises pulsar recall by 5.79 percentage points, but lowers precision and accuracy. This is useful when missing a pulsar is considered more costly than reviewing extra false positives.

Unsupervised models

Method	Our result	Reference paper	Finding
K-Means	3 clusters; silhouette 0.4413	3 clusters; 0.44	Close reproduction
SOM (5×5 map)	25 occupied neurons; silhouette 0.1668	3 occupied neurons; 0.44	Does not match reference yet

- K-Means closely matches the paper's silhouette score. Its cluster counts are 621, 44, and 974 among the 1,639 pulsar samples.
- The SOM uses a 5×5 grid and 100 training epochs in the stated training output. All 25 neurons receive samples, unlike the paper's three non-empty cells. The lower silhouette score also indicates weaker separation under the current evaluation. This difference should be reported as a limitation, not described as an exact reproduction.
- The first two PCA components explain 81.98% of variance, offering a useful two-dimensional view while still omitting some information from the eight original features.

4. Part II — Additional Model Results: KNN

Purpose. KNN extends the reference experiment by testing a different supervised learning strategy. It classifies a sample from the labels of its nearest training examples.

KNN evaluation	Result
Selected neighbour count	k = 3
Five-fold CV accuracy / recall	97.74% / 83.30%
Test accuracy	98.02%
Test precision / recall	92.13% / 85.67%
Test F1-score / ROC-AUC	88.78% / 0.9457

- KNN reaches the same test accuracy as Random Forest at threshold 0.5, but its ROC-AUC is lower (0.9457 versus 0.9661).
- The extension demonstrates that model comparison must consider precision, recall, F1-score, and ROC-AUC—not accuracy alone. KNN adds a useful benchmark but does not outperform all reproduced models.

5. Comparison and Key Findings

Key question	Finding
Did we reproduce the main classification behavior?	Broadly yes; GDA and Random Forest results are close to or above reported values.
Which clustering method most closely matches?	K-Means: 0.4413 versus the paper's 0.44 silhouette score.
What remains different?	SOM: 25 occupied neurons and 0.1668 silhouette versus 3 cells and 0.44 in the paper.
What does KNN add?	An extra supervised baseline with 98.02% test accuracy and a clear multi-metric comparison.

Conclusion. The project completes both required parts: reproducing GDA, Random Forest, K-Means, and SOM, then extending the study with KNN. Classification and K-Means results are reasonably close to the reference, while SOM remains a partial reproduction. The results show why class imbalance, recall, decision thresholds, and multiple evaluation metrics matter in pulsar identification.

Reference

“Application of Machine Learning Methods to Identify and Categorize Radio Pulsar Signal Candidates.”